INFOSYS SE MASTER COMPENDIUM

TOPIC: PERMUTATION, COMBINATION & PROBABILITY

Total Questions: 28 | All Question Variations & Solved Solutions

QUESTION 1

[CampusMonk] 2. Full length Quantitative Aptitude (with sol) Level -2

In an envelope there are 5 green, 3 yellow and 4 pink tablets. 3 tablets are picked at random.Quantity I: The probability that 2 tablets are yellow in colour and 1 tablet is pink in colour.Quantity II: The probability that all the tablets are green in colour.

(A)

(B)

(C)

(D)

(E)

✓ Correct Answer: (A)

Ans - A

QUESTION 2

[CampusMonk] 2. Full length Verbal Ability

Change the voice:
Her selection in the crew surprised Peter.

(A) Her selection in the crew was a big surprise.

(B) Peter was surprised at her selection in the crew.

(C) Her selection was a surprise.

(D) Peter was surprised at her selection of the crew.

✓ Correct Answer: (B)

Ans- b

QUESTION 3

[CampusMonk] 3. Full length Quantitative Aptitude (with sol) Level -1-2

A bag contains 5 red and 7 white balls. Four balls are drawn out one by one and not replaced. What is the probability that they are alternatively of different colours?

(A) 7/99

(B) 11/99

(C) **14/99**

(D) 19/99

(E) None of these

✓ Correct Answer: (C)

Ans - C

QUESTION 4

[OnlineStudy4U] INFOSYS TEST SERIES-1

The
probability of getting a sum of 9 when two dice are thrown is:

(A) **1/12**

(B) 1/9

(C) 5/39

(D) 7/36

✓ Correct Answer: (A)

Combinations: (3,6), (4,5), (5,4), (6,3). Total = 4 outcomes. Probability = 4/36 = 1/9 (Correction → 1/9, not 1/12).

QUESTION 5

[OnlineStudy4U] Infosys Full Length Mock Test-1

Two fair dice are thrown together. What is
the probability of getting a sum of 7?

(A) 1/12

(B) 1/8

(C) **1/6**

(D) 5/36

✓ Correct Answer: (C)

Favourable outcomes: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) — 6 outcomes. Total outcomes = 36. Probability = 6/36 = 1/6.

QUESTION 6

[OnlineStudy4U] Infosys Full Length Mock Test- 2

A team of 3 is to be chosen from 5 developers and 4 testers. In how many ways can the team be formed if it must include at least 1 tester?

(A) 64

(B) 70

(C) **74**

(D) 84

✓ Correct Answer: (C)

Total ways to choose 3 from 9 = C(9,3) = 84. Ways with no tester = C(5,3) = 10. So at least one tester = 84 − 10 = 74.

QUESTION 7

[OnlineStudy4U] Infosys Full Length Mock Test- 2

In how many distinct ways can the letters of the word SERVER be arranged?

(A) 120

(B) **180**

(C) 240

(D) 720

✓ Correct Answer: (B)

SERVER has 6 letters with R and E repeating twice each. Number of distinct arrangements = 6! / (2! × 2!) = 720 / 4 = 180.

QUESTION 8

[OnlineStudy4U] Infosys Full Length Mock Test- 2

A bag contains 4 red and 6 blue balls. Two balls are drawn at random without replacement. What is the probability that both are red?

(A) **2/15**

(B) 1/5

(C) 4/15

(D) 2/9

✓ Correct Answer: (A)

P = C(4,2) / C(10,2) = 6 / 45 = 2/15. Sequentially: (4/10)×(3/9) = 12/90 = 2/15.

QUESTION 9

[OnlineStudy4U] Infosys Full Length Mock Test- 2

Two fair dice are thrown. What is the probability that the sum is 9?

(A) 1/12

(B) **1/9**

(C) 1/8

(D) 1/6

✓ Correct Answer: (B)

Favourable pairs: (3,6), (4,5), (5,4), (6,3) — 4 outcomes. Total outcomes = 36. Probability = 4/36 = 1/9.

QUESTION 10

[OnlineStudy4U] Infosys Full Length Mock Test- 4

If the letters of LOGIC are arranged in alphabetical order, which letter is in the middle?

(A) G

(B) I

(C) **L**

(D) O

✓ Correct Answer: (C)

Alphabetical: C, G, I, L, O. The middle letter is I.

QUESTION 11

[OnlineStudy4U] Infosys Full Length Mock Test- 5

Two cards are drawn from 52 without replacement. Probability both are aces is

(A) **1/221**

(B) 1/169

(C) 1/17

(D) 2/52

✓ Correct Answer: (A)

C(4,2)/C(52,2) = 6/1326 = 1/221.

QUESTION 12

[OnlineStudy4U] Infosys Full Length Mock Test- 5

A fair coin is tossed three times.
Probability of exactly two heads is

(A) 1/8

(B) **3/8**

(C) 1/2

(D) 3/4

✓ Correct Answer: (B)

C(3,2)/8 = 3/8.

QUESTION 13

[OnlineStudy4U] Infosys Full Length Mock Test- 5

A bag has 3 white and 5 black balls. One ball is drawn. Probability it is black is

(A) 3/8

(B) 1/2

(C) **5/8**

(D) 2/5

✓ Correct Answer: (C)

5 black out of 8 = 5/8.

QUESTION 14

[OnlineStudy4U] Infosys Full Length Mock Test- 5

How many ways can 5 distinct books be arranged on a shelf if two particular books must stay together?

(A) **48**

(B) 60

(C) 120

(D) 240

✓ Correct Answer: (A)

Tie the two books as one bundle: 4 units can be arranged in 4! ways and the bundle in 2! ways. 24 × 2 = 48.

QUESTION 15

[OnlineStudy4U] Infosys Full Length Mock Test- 6

Two coins are tossed. Probability of at least one tail is

(A) 1/4

(B) 1/2

(C) **3/4**

(D) 1

✓ Correct Answer: (C)

Outcomes HH, HT, TH, TT. At least one tail = 3/4.

QUESTION 16

[OnlineStudy4U] Infosys Full Length Mock Test- 6

A bag contains 5 green and 3 yellow balls. One ball is drawn at random. Probability it is yellow is

(A) **3/8**

(B) 5/8

(C) 3/5

(D) 1/3

✓ Correct Answer: (A)

3 yellow out of 8 = 3/8.

QUESTION 17

[OnlineStudy4U] Infosys Full Length Mock Test- 6

The letters of the word PAPER can be arranged in how many distinct ways?

(A) **60**

(B) 120

(C) 24

(D) 80

✓ Correct Answer: (A)

PAPER has 5 letters and P repeats twice. 5! / 2! = 60.

QUESTION 18

[OnlineStudy4U] Infosys Full Length Mock Test- 6

A
die is thrown once. What is the probability of a prime number?

(A) 1/3

(B) **1/2**

(C) 2/3

(D) 1/6

✓ Correct Answer: (B)

Primes on a die: 2, 3, 5. Probability = 3/6 = 1/2.

QUESTION 19

[OnlineStudy4U] Infosys Full Length Mock Test- 6

In
how many ways can a team of 4 be chosen from 7 players?

(A) 28

(B) 30

(C) **35**

(D) 84

✓ Correct Answer: (C)

C(7,4) = 35.

QUESTION 20

[OnlineStudy4U] Infosys Full Length Mock Test- 8

A
fair die is thrown. Probability of a number greater than 4 is

(A) 1/6

(B) **1/3**

(C) 1/2

(D) 2/3

✓ Correct Answer: (B)

5 and 6: 2/6 = 1/3.

QUESTION 21

[OnlineStudy4U] Infosys Full Length Mock Test- 8

A
bag has 5 white and 7 black balls. One ball is drawn. Probability
it is white is

(A) 5/7

(B) 1/2

(C) **5/12**

(D) 7/12

✓ Correct Answer: (C)

5 white out of 12 = 5/12.

QUESTION 22

[OnlineStudy4U] Infosys Full Length Mock Test- 8

Two
coins are tossed. Probability of two heads is

(A) **1/4**

(B) 1/2

(C) 1/3

(D) 3/4

✓ Correct Answer: (A)

HH is 1 of 4 outcomes.

QUESTION 23

[OnlineStudy4U] Infosys Full Length Mock Test- 8

The
letters of the word OFFER can be arranged in how many distinct
ways?

(A) **60**

(B) 120

(C) 24

(D) 80

✓ Correct Answer: (A)

5 letters with F twice: 5! / 2! = 60.

QUESTION 24

[OnlineStudy4U] Infosys Full Length Mock Test- 9

The
letters of the word ERROR can be arranged in how many distinct
ways?

(A) 20 | (B) 60 | (C) 120 | (D) 24

✓ Correct Answer: (A)

5 letters, R three times. 5! / 3! = 20.

QUESTION 25

[OnlineStudy4U] Infosys Full Length Mock Test- 9

In
how many ways can 2 students be chosen from 9 students?

(A) 18 | **(B) 36** | (C) 72 | (D) 81

✓ Correct Answer: (B)

C(9,2)=36.

QUESTION 26

[OnlineStudy4U] Infosys Full Length Mock Test- 9

A
card is drawn from 52. Probability it is a queen is

(A) 1/13 | (B) 1/4 | (C) 1/26 | (D) 4/13

✓ Correct Answer: (A)

4 queens in 52 = 1/13.

QUESTION 27

[OnlineStudy4U] Infosys Full Length Mock Test- 9

A
number is picked from 1 to 10. Probability it is even is

(A) 2/5

(B) **1/2**

(C) 3/5

(D) 1/5

✓ Correct Answer: (B)

2,4,6,8,10 → 5/10 = 1/2.

QUESTION 28

[OnlineStudy4U] Infosys Full Length Mock Test- 9

A
coin is tossed twice. Probability of one head and one tail is

(A) 1/4

(B) 1/3

(C) **1/2**

(D) 3/4

✓ Correct Answer: (C)

HT and TH: 2/4 = 1/2.